

LANDSBERG AM LECH, Germany

WMO: 108570

Lat: 48.071N

Lon: 10.906E

Elev: 623

StdP: 94.06

Time Zone: 1.00 (EUC)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.0	-10.1	-14.1	1.2	-12.6	-11.4	1.5	-9.1	15.1	7.5	13.4	6.8	2.0	170	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	29.8	18.8	27.9	18.2	26.1	17.6	19.6	27.4	18.8	26.2	18.1	25.2	2.7	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.9	13.0	22.5	16.1	12.3	21.8	15.2	11.7	21.0	58.5	27.7	55.9	26.4	53.5	25.4	22.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.1	7.6	-17.0	32.7	3.4	1.5	-19.5	33.7	-21.4	34.6	-23.3	35.4	-25.8	36.5 (4)		
(5)			-17.1	21.2	3.4	0.7	-19.6	21.7	-21.6	22.0	-23.5	22.4	-25.9	22.9 (5)		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-0.5	0.5	4.3	8.5	13.0	16.5	18.2	18.0	13.5	9.3	3.9	0.5	(6)
(7)		DBStd	7.86	4.88	5.15	4.37	4.10	3.66	3.75	3.30	3.17	3.50	4.06	4.21	4.57	(7)
(8)		HDD10.0	1412	326	267	179	75	13	1	0	0	8	62	187	295	(8)
(9)		HDD18.3	3564	584	500	434	296	168	78	44	46	148	279	433	553	(9)
(10)		CDD10.0	999	0	1	4	30	107	197	253	247	114	42	4	0	(10)
(11)		CDD18.3	106	0	0	0	0	4	24	39	34	4	0	0	0	(11)
(12)		CDH23.3	1180	0	0	0	6	66	265	417	373	51	3	0	0	(12)
(13)		CDH26.7	292	0	0	0	1	8	60	117	102	4	0	0	0	(13)
(14)	Wind	WSAvg	3.2	3.7	3.7	3.8	3.2	3.1	2.8	2.9	2.7	2.8	2.8	3.2	3.5	(14)
(15)	Precipitation	PrecAvg	947	51	47	62	62	105	116	119	110	79	67	66	62	(15)
(16)		PrecMax	1286	119	101	149	141	213	214	248	220	152	167	151	130	(16)
(17)		PrecMin	705	1	14	19	10	7	50	20	41	23	16	0	2	(17)
(18)		PrecStd	131	30	24	31	34	48	43	49	47	36	34	35	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	15.1	19.0	23.9	28.0	30.8	32.3	32.1	27.1	22.9	17.2	12.4	(19)
(20)			MCWB	7.7	7.9	10.0	13.3	16.8	19.0	19.5	19.1	17.4	14.8	11.4	7.5	(20)
(21)		2%	DB	10.1	12.0	16.0	21.2	25.2	28.5	29.8	29.2	24.9	20.1	14.2	10.1	(21)
(22)			MCWB	6.6	6.7	9.1	12.2	16.1	18.7	18.8	18.8	16.7	14.2	10.0	6.8	(22)
(23)		5%	DB	8.4	9.6	13.8	18.9	22.9	26.5	27.6	27.2	22.7	18.0	12.0	8.6	(23)
(24)			MCWB	5.4	5.6	8.3	11.5	14.9	17.9	18.2	18.0	15.8	12.9	8.7	6.0	(24)
(25)		10%	DB	6.3	7.7	11.3	16.6	20.8	24.4	25.6	25.2	20.2	15.8	10.0	6.9	(25)
(26)			MCWB	4.2	4.7	7.1	10.2	13.9	16.9	17.6	17.5	14.9	12.2	7.5	4.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	8.8	11.1	14.1	18.1	20.6	20.5	20.3	18.2	15.8	11.7	8.5	(27)
(28)			MCDWB	11.6	13.5	16.9	22.0	25.7	27.9	29.3	28.5	25.5	21.0	15.8	11.4	(28)
(29)		2%	WB	7.0	7.1	9.7	12.7	16.7	19.2	19.5	19.4	17.2	14.6	10.3	7.4	(29)
(30)			MCDWB	9.7	10.9	14.7	19.9	23.9	26.7	27.4	27.1	23.4	19.0	13.6	9.7	(30)
(31)		5%	WB	5.5	5.9	8.6	11.7	15.4	18.3	18.7	18.6	16.2	13.5	9.1	6.1	(31)
(32)			MCDWB	7.9	9.2	12.9	18.0	21.7	25.6	26.1	25.9	21.6	17.2	11.7	8.1	(32)
(33)		10%	WB	4.2	4.6	7.4	10.5	14.3	17.2	17.9	17.8	15.2	12.4	7.9	4.9	(33)
(34)			MCDWB	6.1	7.4	11.1	15.9	19.7	23.6	24.7	24.5	19.4	15.4	9.8	6.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.5	7.8	9.0	10.5	10.5	10.4	10.7	10.8	9.7	8.8	6.8	6.2	(35)
(36)			MCDBR	9.1	11.5	13.7	15.3	14.9	14.4	14.9	14.8	14.1	12.5	10.9	8.9	(36)
(37)			MCWBR	6.1	7.2	8.0	7.9	7.4	6.6	6.2	6.4	7.2	7.2	7.2	6.1	(37)
(38)		5% WB	MCDWB	8.4	10.4	12.4	14.0	13.4	13.1	13.1	13.5	12.5	11.2	9.9	8.1	(38)
(39)			MCWBR	6.0	6.8	7.4	7.5	6.9	6.2	5.8	6.1	6.6	6.8	6.8	5.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.310	0.344	0.377	0.388	0.393	0.395	0.389	0.365	0.336	0.308	0.283	(40)	
(41)		taud	2.459	2.396	2.368	2.301	2.308	2.324	2.310	2.345	2.392	2.466	2.498	2.475	(41)	
(42)		Ebn at Noon	800	851	872	873	875	870	862	851	839	810	766	758	(42)	
(43)		Edh at Noon	64	84	102	120	124	123	123	114	99	78	62	56	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.12	2.01	3.12	4.29	4.93	5.52	5.35	4.71	3.42	2.21	1.21	0.91	(44)	
(45)		RadStd	0.14	0.27	0.37	0.57	0.57	0.51	0.49	0.48	0.42	0.26	0.16	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (87 neighbors)	N/A	N/A	N/A	N/A	+0.41	+0.40	N/A	N/A	+59	+20

Nomenclature: See separate page